C

IB MATH: APPLICATIONS & INTERPRETATION • UNIT 1.01

Vectors & Embeddings in AI

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 D	Q2 C	Q3 C	Q4 C	Q5 C
Q6 C	Q7 C	Q8 A	Q9 C	Q10 B
Q11 B	Q12 D	Q13 B	Q14 A	Q15 B
Q16 D	Q17 B	Q18 C	Q19 B	Q20 D
Q21 A	Q22 B	Q23 B	Q24 D	Q25 B

Vectors & Embeddings in AI Practice Exercises

1 MCQ

Compute the Manhattan (L_1) norm and Euclidean (L_2) norm of the word vector $\mathbf{w} = [10, -15]^T$.

A $\|\mathbf{w}\|_1 = \sqrt{325},\ \|\mathbf{w}\|_2 = 25$

B $\|\mathbf{w}\|_1 = -5,\ \|\mathbf{w}\|_2 = 325$

C $\|\mathbf{w}\|_1 = 25,\ \|\mathbf{w}\|_2 = 25$

D $\|\mathbf{w}\|_1 = 25,\ \|\mathbf{w}\|_2 = \sqrt{325}$ ✓

WORKED SOLUTION & VERIFICATION

L_1 norm is the sum of absolute values: $|10| + |-15| = 25$. L_2 norm is the square root of sum of squares: $\sqrt{(10)^2 + (-15)^2} = \sqrt{100 + 225} = \sqrt{325}$.

2 MCQ

Two document embeddings $\mathbf{a} = [1, 2]^T$ and $\mathbf{b} = [2, 1]^T$ are compared. If $\|\mathbf{a}\| = \|\mathbf{b}\| = \sqrt{5}$, calculate their exact cosine similarity.

A 1.000

B 0.640

C 0.800 ✓

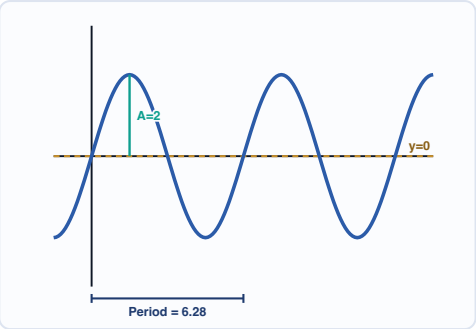
D 0.500

WORKED SOLUTION & VERIFICATION

Dot product $\mathbf{a} \cdot \mathbf{b} = (1)(2) + (2)(1) = 4$. Product of norms $\|\mathbf{a}\|\|\mathbf{b}\| = 5$. Cosine similarity = $\frac{4}{5} = 0.800$.

3 MCQ

Two document embeddings $\mathbf{a} = [2, 4]^T$ and $\mathbf{b} = [4, 2]^T$ are compared. If $\|\mathbf{a}\| = \|\mathbf{b}\| = \sqrt{20}$, calculate their exact cosine similarity.



A 1.000

B 0.640

C 0.800 ✓

D 0.500

WORKED SOLUTION & VERIFICATION

Dot product $\mathbf{a} \cdot \mathbf{b} = (2)(4) + (4)(2) = 16$. Product of norms $\|\mathbf{a}\|\|\mathbf{b}\| = 20$. Cosine similarity = $\frac{16}{20} = 0.800$.

4 MCQ

Two document embeddings $\mathbf{a} = [3, 6]^T$ and $\mathbf{b} = [6, 3]^T$ are compared. If $\|\mathbf{a}\| = \|\mathbf{b}\| = \sqrt{45}$, calculate their exact cosine similarity.

A 1.000

B 0.640

C 0.800 ✓

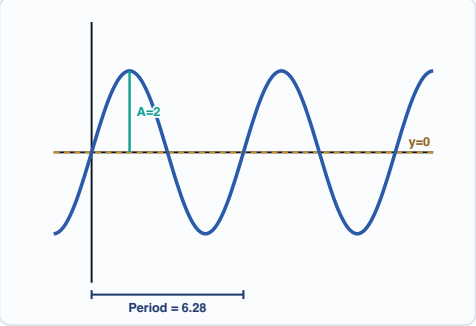
D 0.500

WORKED SOLUTION & VERIFICATION

Dot product $\mathbf{a} \cdot \mathbf{b} = (3)(6) + (6)(3) = 36$. Product of norms $\|\mathbf{a}\|\|\mathbf{b}\| = 45$. Cosine similarity = $\frac{36}{45} = 0.800$.

5 MCQ

Two document embeddings $\mathbf{a} = [4, 8]^T$ and $\mathbf{b} = [8, 4]^T$ are compared. If $\|\mathbf{a}\| = \|\mathbf{b}\| = \sqrt{80}$, calculate their exact cosine similarity.



A 1.000

B 0.640

C 0.800 ✓

D 0.500

WORKED SOLUTION & VERIFICATION

Dot product $\mathbf{a} \cdot \mathbf{b} = (4)(8) + (8)(4) = 64$. Product of norms $\|\mathbf{a}\|\|\mathbf{b}\| = 80$. Cosine similarity = $\frac{64}{80} = 0.800$.

6 MCQ

Two document embeddings $\mathbf{a} = [5, 10]^T$ and $\mathbf{b} = [10, 5]^T$ are compared. If $\|\mathbf{a}\| = \|\mathbf{b}\| = \sqrt{125}$, calculate their exact cosine similarity.

A 1.000	B 0.640
C 0.800 ✓	D 0.500

WORKED SOLUTION & VERIFICATION

Dot product $\mathbf{a} \cdot \mathbf{b} = (5)(10) + (10)(5) = 100$. Product of norms $\|\mathbf{a}\|\|\mathbf{b}\| = 125$. Cosine similarity = $\frac{100}{125} = 0.800$.

7 MCQ

Two document embeddings $\mathbf{a} = [6, 12]^T$ and $\mathbf{b} = [12, 6]^T$ are compared. If $\|\mathbf{a}\| = \|\mathbf{b}\| = \sqrt{180}$, calculate their exact cosine similarity.

A 1.000	B 0.640
C 0.800 ✓	D 0.500

WORKED SOLUTION & VERIFICATION

Dot product $\mathbf{a} \cdot \mathbf{b} = (6)(12) + (12)(6) = 144$. Product of norms $\|\mathbf{a}\|\|\mathbf{b}\| = 180$. Cosine similarity = $\frac{144}{180} = 0.800$.

8 MCQ

The coordinate diagram shows the parabolic curve $f_1(x) = x^2 - 6x + 20$ in the domain of Vectors & Embeddings in AI. By completing the square $f_1(x) = (x - h)^2 + k$, find the sum $k + f_1(5)$.

A 26 ✓	B 29
C 15	D 24

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Vectors & Embeddings in AI):

1. Complete the square: $x^2 - 6x + 20 = (x - 3)^2 - 9 + 20 = (x - 3)^2 + 11$, giving minimum value $k = 11$. **[M1]**
2. Evaluate at $x = 5$: $f_1(5) = (2)^2 + 11 = 15$. **[M1]**
3. Sum the two quantities: $k + f_1(5) = 11 + 15 = 26$. **[A1]**

- **Final Verified Answer:** 26.

9 MCQ

Consider two vectors, $\mathbf{u} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$. What is the vector representing $2\mathbf{u} - \mathbf{v}$?

A $\begin{bmatrix} 2 \\ 12 \end{bmatrix}$

B $\begin{bmatrix} 5 \\ 6 \end{bmatrix}$

C $\begin{bmatrix} 5 \\ 10 \end{bmatrix}$ ✓

D $\begin{bmatrix} 7 \\ 10 \end{bmatrix}$

WORKED SOLUTION & VERIFICATION

To find the vector $2\mathbf{u} - \mathbf{v}$, we first compute $2\mathbf{u}$ and then subtract $\mathbf{v}$ from the result.

$2\mathbf{u} = 2 \times \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 2 \times 3 \\ 2 \times 4 \end{bmatrix} = \begin{bmatrix} 6 \\ 8 \end{bmatrix}.$

Now, we subtract $\mathbf{v}$ from $2\mathbf{u}$:

$2\mathbf{u} - \mathbf{v} = \begin{bmatrix} 6 \\ 8 \end{bmatrix} - \begin{bmatrix} 1 \\ -2 \end{bmatrix} = \begin{bmatrix} 6 - 1 \\ 8 - (-2) \end{bmatrix} = \begin{bmatrix} 5 \\ 8 + 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 10 \end{bmatrix}.$

Therefore, the resulting vector is $\begin{bmatrix} 5 \\ 10 \end{bmatrix}$.

10 MCQ

A vector $\mathbf{v}$ is transformed into a new vector $\mathbf{v}'$ by a linear transformation represented by the matrix $\mathbf{M}$. We are given that $\mathbf{v} = (2\ 3)$ and $\mathbf{v}' = (7\ 11)$. The transformation matrix is of the form $\mathbf{M} = (a \quad 1\ 4 \quad b)$. Determine the value of a and b.

A a = 3, b = 2

B **a = 2, b = 1** ✓

C a = 4, b = -1

D a = 1, b = 3

WORKED SOLUTION & VERIFICATION

The transformation is defined by $\mathbf{v}' = \mathbf{M}\mathbf{v}$. Substituting the given vectors and matrix form:

$(7\ 11) = (a \quad 1\ 4 \quad b)\,(2\ 3)$

Performing the matrix multiplication:

$(7\ 11) = (2a + 3\ 8 + 3b)$

Equating the corresponding components, we get a system of two linear equations:

1) $2a + 3 = 7$

2) $8 + 3b = 11$

Solving the first equation for a:

$2a = 7 - 3$

$2a = 4$

$a = 2$

Solving the second equation for b:

$3b = 11 - 8$

$3b = 3$

$b = 1$

Thus, $a = 2$ and $b = 1$.

11 MCQ

Examining a dataset, the prior probability of class C_1 is $P(C_1) = 0.60$. What is the probability of the complement class $P(\neg C_1)$?

A 0.70	B 0.40 ✓
C 0.30	D 1.60

WORKED SOLUTION & VERIFICATION

By the complement rule of probability: $P(\neg C_1) = 1 - P(C_1) = 1 - 0.60 = 0.40$.

12 MCQ

Principal Component Analysis on covariance matrix Σ_X yields eigenvalues $\lambda_1 = 83$ and $\lambda_2 = 17$. Determine the percentage of variance h_u explained by the first principal component.

A 87	B 63
C 88	D **83** ✓

WORKED SOLUTION & VERIFICATION

The percentage explained is $\frac{83}{83+17} \times 100\% = 83\%$.

13 MCQ

Consider a dataset of n data points, where each data point is a vector in R^D. A non-linear dimensionality reduction technique maps these data points to a lower-dimensional space R^d. The mapping function for a single data point $\mathbf{x} \in \mathrm{R}^D$ is given by $f(\mathbf{x}) = \phi(\mathbf{x})$, where $\phi : \mathrm{R}^D \to \mathrm{R}^d$ is a function designed to preserve local neighborhood structures. Suppose the mapping ϕ is defined by an explicit feature map $\phi(\mathbf{x}) = [\mathbf{x}^T\mathbf{x}, \mathbf{x}^T\mathbf{x}, \ldots, \mathbf{x}^T\mathbf{x}]^T$ where the vector $\mathbf{x} \in \mathrm{R}^{10}$ and d is the dimension of the embedded space. If the goal is to embed these points into a space such that pairwise Euclidean distances are preserved as closely as possible under a specific distortion, and it is found that $d = 15$ yields optimal results for this mapping under a certain constraint, determine the effective rank of the feature map's image space.

A 15	B **1** ✓
C 10	D 25

WORKED SOLUTION & VERIFICATION

The feature map is defined as $\phi(\mathbf{x}) = [\mathbf{x}^T\mathbf{x}, \mathbf{x}^T\mathbf{x}, \ldots, \mathbf{x}^T\mathbf{x}]^T$, where $\mathbf{x} \in \mathrm{R}^{10}$. The term $\mathbf{x}^T\mathbf{x}$ is the squared Euclidean norm of $\mathbf{x}$, denoted as $\|\mathbf{x}\|^2$. This is a scalar value. Thus, the mapping ϕ produces a vector where each component is the same scalar value, $\|\mathbf{x}\|^2$. Let $k = \|\mathbf{x}\|^2$. Then $\phi(\mathbf{x}) = [k, k, \ldots, k]^T$ in R^d. This means that the image of the mapping ϕ is a subspace spanned by a single vector. For example, if $d = 15$, then $\phi(\mathbf{x}) = [k, k, \ldots, k]^T$ which can be written as $k \cdot [1, 1, \ldots, 1]^T$. The vector $[1, 1, \ldots, 1]^T$ forms a basis for the image space. Therefore, the dimension of the image space, which is equivalent to its rank, is 1, regardless of the value of d (as long as $d \geq 1$). The problem states that $d = 15$, but the effective rank of the image of this specific feature map is always 1, as all output vectors lie on a single line passing through the origin.

14 MCQ

A binary classifier $h(u)$ produces the confusion matrix below with $\text{TP} = 60$, $\text{FP} = 12$, $\text{FN} = 6$, and $\text{TN} = 10$. Determine the total number of correct classifications $k = \text{TP} + \text{TN}$.

- **A** 70 ✓
- **B** 64
- **C** 74
- **D** 85

WORKED SOLUTION & VERIFICATION

Summing $\text{TP} + \text{TN} = 60 + 10 = 70$.

15 MCQ

Consider the scalar loss function $L(w) = 3w^2 - 8w + 5$. Find the analytical gradient $\nabla L(w)$ evaluated at $w = 4$.

- **A** $\nabla L(4) = 24$
- **B** $\nabla L(4) = 16$ ✓
- **C** $\nabla L(4) = 32$
- **D** $\nabla L(4) = 4$

WORKED SOLUTION & VERIFICATION

Differentiate with respect to w: $\frac{dL}{dw} = 2(3)w - 8 = 6w - 8$. At $w = 4$: $\nabla L = 6(4) - 8 = 24 - 8 = 16$.

16 MCQ

Suppose a machine learning diagnostic test has sensitivity $P(\text{Positive}|\text{Defect}) = 0.90$. If the defect rate in production is $P(\text{Defect}) = 0.20$, find the joint probability of a defect and a positive test result $P(\text{Defect} \cap \text{Positive})$.

- **A** 1.1000
- **B** 0.7000
- **C** 0.0200
- **D** 0.1800 ✓

WORKED SOLUTION & VERIFICATION

By the multiplication rule of conditional probability: $P(\text{Defect} \cap \text{Positive}) = P(\text{Defect}) \times P(\text{Positive}|\text{Defect}) = 0.20 \times 0.90 = 0.1800$.

17 MCQ

Suppose a 3D dataset has covariance matrix eigenvalues $\lambda_1 = 68$, $\lambda_2 = 21$, and $\lambda_3 = 11$. What percentage of total variance is explained by the first principal component PC_1?

- **A** 89.0%
- **B** 68.0% ✓
- **C** 21.0%
- **D** 76.4%

WORKED SOLUTION & VERIFICATION

Total variance is $\sum \lambda_i = 68 + 21 + 11 = 100$. The proportion explained by PC_1 is $\frac{\lambda_1}{\sum \lambda_i} = \frac{68}{100} = 68.0\%$.

18 MCQ

Consider the sigmoid activation function is $\sigma(z) = \frac{1}{1+e^{-z}}$. If $\sigma(z_0) = 1.20$, what is the value of the derivative $\sigma'(z_0)$?

A -0.2000

B 1.4400

C -0.2400 ✓

D 2.4000

WORKED SOLUTION & VERIFICATION

The derivative of sigmoid satisfies $\sigma'(z) = \sigma(z)(1-\sigma(z))$. Substituting $\sigma(z_0) = 1.20$: $\sigma'(z_0) = (1.20)(1-1.20) = (1.20)(-0.20) = -0.2400$.

19 MCQ

Considering anomaly detection, anomaly prior $P(A) = 0.04$, detection sensitivity $P(+|A) = 0.95$, and false alarm rate $P(+|\neg A) = 0.05$. By Bayes' theorem, what is the posterior probability $P(A|+)$ that an alarm is a true anomaly?

A 0.221

B 0.442 ✓

C 0.950

D 0.619

WORKED SOLUTION & VERIFICATION

By Bayes' Rule: $P(A|+) = \frac{P(+|A)P(A)}{P(+|A)P(A)+P(+|\neg A)P(\neg A)} = \frac{0.95\times0.04}{0.95(0.04)+0.05(0.96)} = \frac{0.0380}{0.0860} = 0.442$.

20 MCQ

For a bivariate feature dataset, $\mathrm{Cov}(x,y) = 72$, $\mathrm{Var}(x) = 24$, $\bar{x} = 5$, and $\bar{y} = 50$. Compute the OLS regression slope $\hat{\beta}_1$ and intercept $\hat{\beta}_0$.

A $\hat{\beta}_1 = 3,\ \hat{\beta}_0 = 50$

B $\hat{\beta}_1 = 4,\ \hat{\beta}_0 = 30$

C $\hat{\beta}_1 = \frac{1}{3},\ \hat{\beta}_0 = 35$

D $\hat{\beta}_1 = 3,\ \hat{\beta}_0 = 35$ ✓

WORKED SOLUTION & VERIFICATION

The OLS slope is $\hat{\beta}_1 = \frac{\mathrm{Cov}(x,y)}{\mathrm{Var}(x)} = \frac{72}{24} = 3$. The intercept is $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x} = 50 - (3)(5) = 50 - 15 = 35$.

21 MCQ

A centered feature matrix $\mathbf{X}$ has shape $N \times 18$. What is the size and symmetry property of the sample covariance matrix $\mathbf{\Sigma} = \frac{1}{N-1}\mathbf{X}^T\mathbf{X}$?

A 18×18 **symmetric matrix** ✓

B $N \times N$ symmetric matrix

C 18×18 skew-symmetric matrix

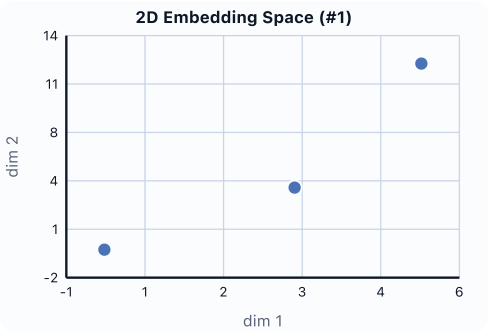
D $N \times 18$ rectangular matrix

WORKED SOLUTION & VERIFICATION

Since $\mathbf{X}$ has 18 columns, $\mathbf{X}^T$ is $18 \times N$ and $\mathbf{X}^T\mathbf{X}$ is 18×18. Furthermore, $(\mathbf{X}^T\mathbf{X})^T = \mathbf{X}^T(\mathbf{X}^T)^T = \mathbf{X}^T\mathbf{X}$, proving it is always symmetric.

22 MCQ

Compute the cosine similarity $\cos(\theta) = \frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{u}\|_2\|\mathbf{v}\|_2}$ between embedding vectors $\mathbf{u} = [3, 4]^\top$ and $\mathbf{v} = [5, 12]^\top$ (embedding pair #1).



A $\frac{66}{65}$

B $\frac{63}{65}$ ✓

C $\frac{4}{9}$

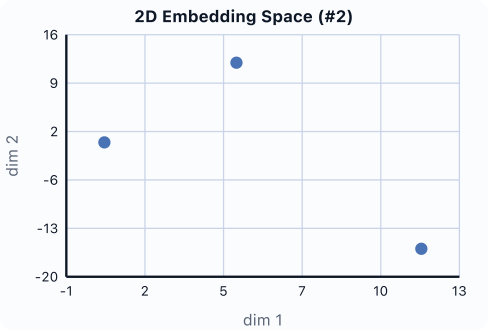
D $\frac{63}{130}$

WORKED SOLUTION & VERIFICATION

Here $\mathbf{u}\cdot\mathbf{v} = 63$, $\|\mathbf{u}\|_2 = 5$, and $\|\mathbf{v}\|_2 = 13$, giving $\cos(\theta) = \frac{63}{65}$.

23 MCQ

Compute the cosine similarity $\cos(\theta) = \frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{u}\|_2\|\mathbf{v}\|_2}$ between embedding vectors $\mathbf{u} = [5, 12]^\top$ and $\mathbf{v} = [12, -16]^\top$ (embedding pair #2).



A $-\frac{33}{130}$

B $-\frac{33}{65}$ ✓

C $\frac{11}{23}$

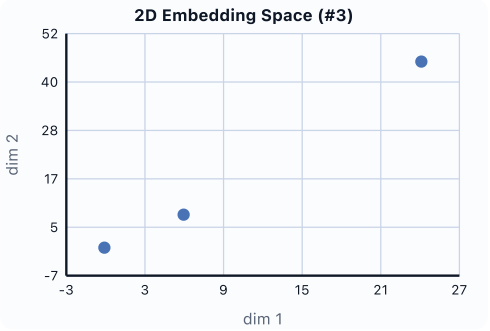
D $\frac{69}{130}$

WORKED SOLUTION & VERIFICATION

Here $\mathbf{u}\cdot\mathbf{v} = -132$, $\|\mathbf{u}\|_2 = 13$, and $\|\mathbf{v}\|_2 = 20$, giving $\cos(\theta) = -\frac{33}{65}$.

24 MCQ

Compute the cosine similarity $\cos(\theta) = \frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{u}\|_2\|\mathbf{v}\|_2}$ between embedding vectors $\mathbf{u} = [6, 8]^\top$ and $\mathbf{v} = [24, 45]^\top$ (embedding pair #3).



A $\frac{14}{27}$

B $\frac{42}{85}$

C $\frac{171}{170}$

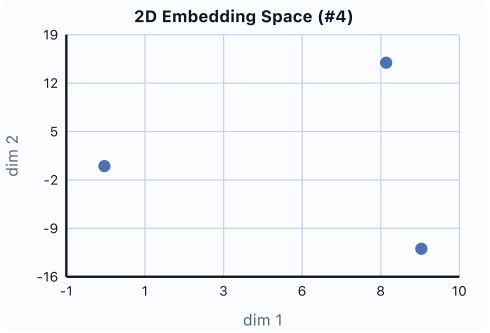
D $\frac{84}{85}$ ✓

WORKED SOLUTION & VERIFICATION

Here $\mathbf{u}\cdot\mathbf{v} = 504$, $\|\mathbf{u}\|_2 = 10$, and $\|\mathbf{v}\|_2 = 51$, giving $\cos(\theta) = \frac{84}{85}$.

25 MCQ

Compute the cosine similarity $\cos(\theta) = \frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{u}\|_2\|\mathbf{v}\|_2}$ between embedding vectors $\mathbf{u} = [8, 15]^\top$ and $\mathbf{v} = [9, -12]^\top$ (embedding pair #4).



A $\frac{37}{85}$

B $-\frac{36}{85}$ ✓

C $-\frac{18}{85}$

D $\frac{17}{32}$

WORKED SOLUTION & VERIFICATION

Here $\mathbf{u}\cdot\mathbf{v} = -108$, $\|\mathbf{u}\|_2 = 17$, and $\|\mathbf{v}\|_2 = 15$, giving $\cos(\theta) = -\frac{36}{85}$.